MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Day-2: File Ownership and Access Control

Reg. Number	Name	Branch
261100690010	Hegde Aansh Dilip	Cyber Security

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Day-2: File Ownership and Access Control

Part A — Ownership Investigation (10 Minutes)

Create a directory named ownership_lab.

```
┌──(kali㉿kali)-[~]
└─$ mkdir ownership_lab
```

Inside it, create: incident_report.txt, network.log, credentials.txt

```
┌──(kali㉿kali)-[~]
└─$ cd ownership_lab

┌──(kali㉿kali)-[~/ownership_lab]
└─$ touch incident_report.txt network.log credentials.txt
```

Display the detailed information for these files.

For each file, identify: Owner, Group, Owner permissions, Group permissions, Others permissions

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ ls -l credentials.txt incident_report.txt network.log
-rw-rw-r-- 1 kali kali 0 Sep  3 13:04 credentials.txt
-rw-rw-r-- 1 kali kali 0 Sep  3 13:04 incident_report.txt
-rw-rw-r-- 1 kali kali 0 Sep  3 13:04 network.log
```

Determine the UID and GID of the current user.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ id
uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),
44(video),46(plugdev),100(users),101(netdev),102(scanner),118(wireshark),119(kaboxer),968(vboxsf),982(bluetooth),99
9(lpadmin)
```

Compare the UID/GID information with the ownership displayed for the newly created files.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ stat incident_report.txt
  File: incident_report.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: 8,1     Inode: 833947      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/    kali)   Gid: ( 1000/    kali)
Access: 2026-09-03 13:04:09.314518934 -0400
Modify: 2026-09-03 13:04:09.314518934 -0400
Change: 2026-09-03 13:04:09.314518934 -0400
 Birth: 2026-09-03 13:04:09.314518934 -0400
```

Explain the difference between file ownership and file permissions.

- **File Ownership:** Defines **WHO** is associated with the file (the specific user owner and group owner).
- **File Permissions:** Defines **WHAT** actions (Read r, Write w, Execute x) can be performed on the file by the Owner, Group, and Others.

Part B — chown: Changing Ownership (15 Minutes)

Change the owner of incident_report.txt to alice.

Change the owner of network.log to bob.

Change the owner of credentials.txt to charlie.

Verify each change.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chown alice incident_report.txt
[sudo] password for kali:

┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chown bob network.log

┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chown charlie credentials.txt

┌──(kali㉿kali)-[~/ownership_lab]
└─$ ls -l
total 0
-rw-rw-r-- 1 charlie kali 0 Sep  3 13:04 credentials.txt
-rw-rw-r-- 1 alice   kali 0 Sep  3 13:04 incident_report.txt
-rw-rw-r-- 1 bob     kali 0 Sep  3 13:04 network.log
```

Try changing the owner of a file to another user without using sudo. Record the result.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ chown bob incident_report.txt
chown: changing ownership of 'incident_report.txt': Operation not permitted
```

Explain why an ordinary user is generally not permitted to transfer file ownership to another user.

- **Disk Quota Bypassing:** Users could transfer massive files to another user's account to avoid exceeding their own storage quotas.
- **Security & Accountability Laundering:** A user could plant malicious files or illegal content on the system and assign ownership to someone else to frame them or evade responsibility.

Using a single command, configure: incident_report.txt, Owner : alice, Group : security

Verify the resulting ownership.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chown alice:security incident_report.txt

┌──(kali㉿kali)-[~/ownership_lab]
└─$ ls -l incident_report.txt
-rw-rw-r-- 1 alice security 0 Sep  3 13:04 incident_report.txt
```

Part C — chgrp: Changing Group Ownership (15 Minutes)

Change the group ownership of network.log to developers.

Change the group ownership of credentials.txt to security.

Verify the changes.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chgrp developers network.log

┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chgrp security credentials.txt

┌──(kali㉿kali)-[~/ownership_lab]
└─$ ls -l network.log credentials.txt
-rw-rw-r-- 1 charlie security   0 Sep  3 13:04 credentials.txt
-rw-rw-r-- 1 bob     developers 0 Sep  3 13:04 network.log
```

Display the group memberships of alice, bob, and charlie.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ id alice
uid=1004(alice) gid=1004(alice) groups=1004(alice),1005(security)

┌──(kali㉿kali)-[~/ownership_lab]
└─$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob),1005(security),1006(developers)

┌──(kali㉿kali)-[~/ownership_lab]
└─$ id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),1005(security)
```

Log in as alice and create a file named alice_report.txt.

Attempt to change the group of alice_report.txt to security.

Attempt to change its group to developers.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ su - alice
Password:
$ cd /path/to/ownership_lab
-sh: 1: cd: can't cd to /path/to/ownership_lab
$ touch alice_report.txt
$ chgrp security alice_report.txt
$ chgrp developers alice_report.txt
chgrp: changing group of 'alice_report.txt': Operation not permitted
$
```

Explain how a user's group membership affects their ability to change the group ownership of a file they own.

An unprivileged user who owns a file can only change the file's group ownership to a group that they currently belong to. They cannot assign a file to a group of which they are not a member.

Part D — Ownership + Permission Investigation (20 Minutes)

Configure:

-rw-r----- alice security incident_report.txt

Identify what operations are available to: alice

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chown alice:security incident_report.txt
[sudo] password for kali:

┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chmod 640 incident_report.txt

┌──(kali㉿kali)-[~/ownership_lab]
└─$ ls -l incident_report.txt
-rw-r----- 1 alice security 0 Sep  3 13:04 incident_report.txt
```

A user who belongs to neither security nor the file owner

charlie belongs to security. Test whether Charlie can:

Read incident_report.txt

Modify incident_report.txt

Delete incident_report.txt

Record the result of each operation.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chown alice:security incident_report.txt
[sudo] password for kali:

┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chmod 640 incident_report.txt

┌──(kali㉿kali)-[~/ownership_lab]
└─$ ls -l incident_report.txt
-rw-r----- 1 alice security 0 Sep  3 13:04 incident_report.txt

┌──(kali㉿kali)-[~/ownership_lab]
└─$ su - charlie
Password:
$ cd ~/ownership_lab
-sh: 1: cd: can't cd to /home/charlie/ownership_lab
$ cat incident_report.txt
cat: incident_report.txt: No such file or directory
$ echo "Charlie log entry" >> incident_report.txt
$ rm incident_report.txt
$ exit
```

Change the permissions to: -rw-rw----

Repeat the test as charlie.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo chmod 660 incident_report.txt

┌──(kali㉿kali)-[~/ownership_lab]
└─$ su - charlie
Password:
$ cd ~/ownership_lab
-sh: 1: cd: can't cd to /home/charlie/ownership_lab
$ cat incident_report.txt
cat: incident_report.txt: No such file or directory
$ echo "Charlie log entry" >> incident_report.txt
$ exit
```

Explain why Charlie's access changed even though the owner and group of the file did not change.

Linux evaluates permissions sequentially by User, Group, then Others. Changing permissions to 660 granted write access to the security group, which automatically gave Charlie write privileges as a group member without altering file ownership.

Remove bob from the security group.

Without changing the file's ownership or permissions, test Bob's access again.

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ sudo gpasswd -d bob security
Removing user bob from group security
```

```
┌──(kali㉿kali)-[~/ownership_lab]
└─$ su - bob
Password:
$ cd ~/ownership_lab
-sh: 1: cd: can't cd to /home/bob/ownership_lab
$ cat incident_report.txt
cat: incident_report.txt: No such file or directory
$ exit
```

Explain why Bob's effective permissions changed.

Linux evaluates access in strict order: Owner, Group, then Others. Removing Bob from the security group caused Linux to bypass Group permissions and fall back to Others (---), revoking his access without changing the file.